Project Argus: Mitigating Stealth Anomalies in Air-Gapped Cyber-Physical Systems via Lightweight Edge AI

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Domain Focus: Applied Deep Learning, Cyber-Physical Systems, Defense R&D

Abstract

Project Argus was developed to address a critical vulnerability in the intersection of Artificial Intelligence and safety-critical infrastructure (e.g., nuclear power, defense systems, and water treatment). Because these Industrial Control Systems (ICS) are air-gapped, standard cloud-reliant AI is non-viable. This paper documents the phased development of a lightweight, domain-agnostic Edge AI framework. By identifying and mathematically mitigating the "Low-Entropy Blind Spot" inherent in standard deep learning models, Project Argus successfully defends against sensor-freezing Replay Attacks. The architecture was validated against the academic HAI 22.04 dataset, proving its scalability from synthetic prototypes to 174-dimensional real-world telemetry.

Introduction

The catalyst for Project Argus was an investigation into the cybersecurity of thermodynamic systems, specifically within the AI x Nuclear domain. Modern ICS architectures rely heavily on Programmable Logic Controllers (PLCs) and Supervisory Control and Data Acquisition (SCADA) networks.

While traditional security thresholds easily catch brute-force sabotage (e.g., a sudden temperature spike), sophisticated state-sponsored actors utilize stealth tactics. The primary threat identified for this research was the **Replay Attack**: a scenario where an adversary compromises a sensor, freezing its output to broadcast a perfectly normal, static baseline to the control room, while physically sabotaging the machinery in reality. The engineering goal was established: build a localized, Edge-deployable AI capable of catching these stealth attacks without human intervention.

Baseline Architecture & The "Blind Spot"

The initial development phase involved constructing an industry-standard Long Short-Term Memory Autoencoder (LSTM-AE). LSTMs are uniquely suited for sequential telemetry because they retain temporal memory.

During synthetic testing, however, a critical flaw in the standard literature was uncovered. When fed data simulating a Replay Attack (a completely flat, frozen temperature reading), the AI failed to flag an anomaly.

- **The Cause:** Autoencoders detect anomalies by measuring the Mean Squared Error (*MSE*) between the input sequence and its reconstruction. Because a flatline is mathematically devoid of complexity, the neural network reconstructed it perfectly, yielding a near-zero *MSE*.
- **The Conclusion:** Standard time-series AI possesses a "Low-Entropy Blind Spot." It inherently trusts perfectly static data, essentially clearing the cyber-attack as safe.

Algorithmic Innovation

To resolve this vulnerability, standard practices suggest exponentially increasing the neural network's parameter size to force it to learn the difference between "natural noise" and "artificial silence." However, this violated the project's core constraint: the model had to remain lightweight enough for Edge microprocessor deployment.

The breakthrough involved shifting the computational burden from the Deep Learning layer to a deterministic Feature Engineering layer. A **Multivariate Rolling Variance** algorithm was introduced prior to tensor sequence creation. For every sensor, the system actively calculates its volatility over a moving temporal window:

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \bar{x})^2$$

This solved the Replay Attack purely mathematically. When a sensor is spoofed and flatlines, its variance drops precisely to 0.0. The lightweight neural network (~3,300 parameters) was retrained to recognize that 0.0 variance in a high-pressure thermodynamic system is a physical impossibility, immediately triggering a massive anomaly spike.

Empirical Validation

A prototype built on synthetic data is insufficient for advanced R&D. The architecture required validation against a rigorous academic benchmark. The codebase was completely refactored into a "Domain-Agnostic Pipeline," meaning it no longer relied on hardcoded sensor names, but instead dynamically analyzed the shape of any incoming data matrix.

The system was tested against the **HIL-based Augmented ICS (HAI 22.04) Security Dataset**, developed by researchers at the Affiliated Institute of ETRI. The HAI 22.04 dataset provides

telemetry from a complex, hardware-in-the-loop simulated steam-turbine and pumped-storage hydropower system.

- **The Result:** The pipeline successfully ingested 87 raw sensor streams, engineered an additional 87 variance features on the fly, and built a custom 174-dimensional LSTM brain. Upon inference against the test data (`test1.csv`), Project Argus successfully identified the highly sophisticated cyber-attacks hidden by the researchers, proving its real-world viability.

Edge Deployment and UI Optimization

The final phase translated the backend Python architecture into a deployable diagnostic dashboard. During integration, a major systems engineering bottleneck occurred: attempting to render 174 overlapping, high-frequency sensor streams simultaneously exceeded standard browser memory limits (the 200MB threshold), causing catastrophic UI failure.

This was resolved by engineering a dynamic, multi-select rendering protocol within the Streamlit interface. While the Edge AI engine continuously calculates the *MSE* loss for all 174 dimensions in the background, the UI defaults to plotting only the top two sensors alongside the Anomaly Probability curve. This guarantees a clean, operator-friendly interface without sacrificing underlying mathematical security.

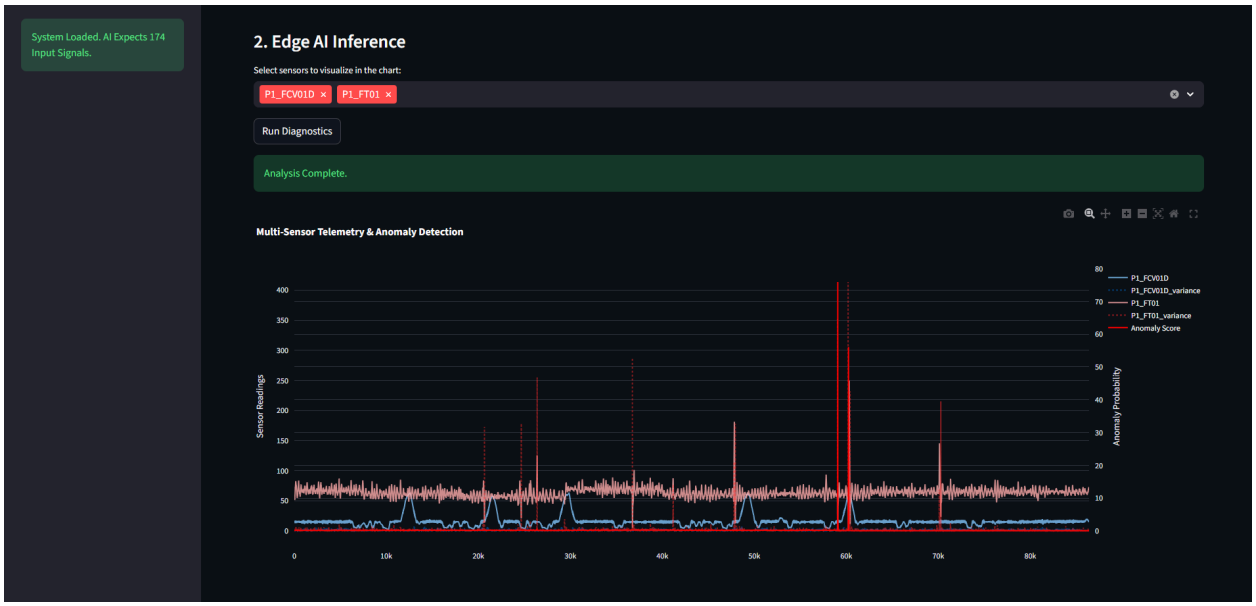

Figure 1: The Argus Edge Inference Dashboard processing the 174-dimensional HAI 22.04 telemetry dataset. Note the critical Anomaly Probability spike (red) at t=60,000, successfully identifying the stealth cyber-attack while the underlying physical sensor readings remained deceptively stable.

Future Scope & Defense R&D Pathway

Project Argus successfully establishes a foundational Proof of Concept for Edge-deployed ICS defense. Future iterations will focus on:

1. **Probabilistic Upgrades:** Transitioning the deterministic LSTM-AE to a Variational Autoencoder (VAE) to provide operators with statistical confidence distributions.
2. **FDIA Mitigation:** Integrating cross-correlation matrices to defend against False Data Injection Attacks, where multiple sensors are subtly altered in tandem.
3. **Hardware Porting:** Optimizing the model via 8-bit quantization for native deployment on microcontrollers, aligning with long-term goals of translating this research into deployable products for national defense and strategic sectors.

References

1. **Malhotra, P., et al. (2015).** *Long Short Term Memory Networks for Anomaly Detection in Time Series.* (Used for the foundational LSTM Autoencoder architecture).
2. **Kravchik, M., & Shabtai, A. (2018).** *Detecting Cyber Attacks in Industrial Control Systems Using Neural Networks.* (Used for the mathematical logic of adding variance to detect stealthy ICS attacks).
3. **An, J., & Cho, S. (2015).** *Variational Autoencoder based Anomaly Detection using Reconstruction Probability.* (Cited for the "Future Work" section regarding the transition to VAEs).